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(54) **METHOD FOR WELL STIMULATION USING NANOBUBBLES**

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(71) Applicant: **Saudi Arabian Oil Company**, Dhahran (SA)

(72) Inventors: **Muhammad Majid Almajid**, Qatif (SA); **Amr Abdel-Fattah**, Dhahran (SA); **Waheed Syed Arshad**, Mississauga (CA); **Afnan Mashat**, Dhahran (SA)

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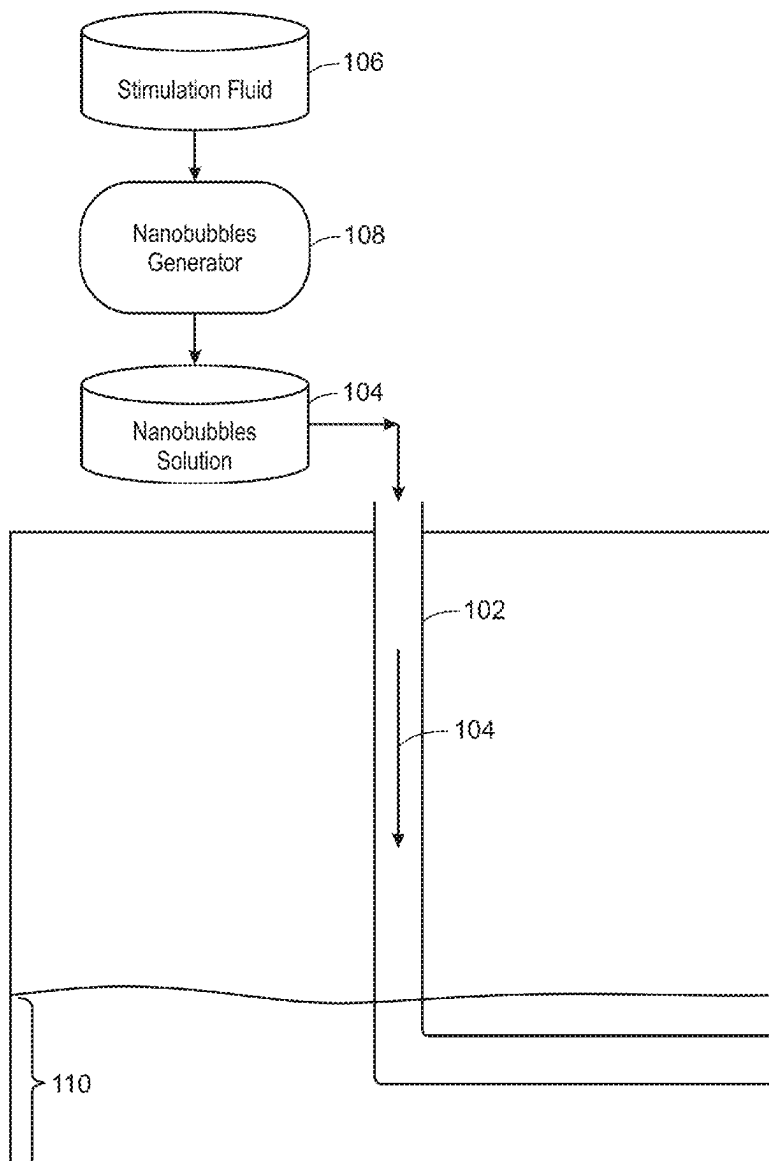
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(62) Division of application No. 18/464,455, filed on Sep. 11, 2023, now Pat. No. 12,352,146.

(57) **ABSTRACT**

A composition and methods are provided for stimulating a well with nanobubbles. An exemplary method includes obtaining a stimulation fluid and generating a nanobubbles solution, wherein the nanobubbles solution includes nano-sized bubbles in the stimulation fluid. The nanobubbles solution is injected into the oil well.



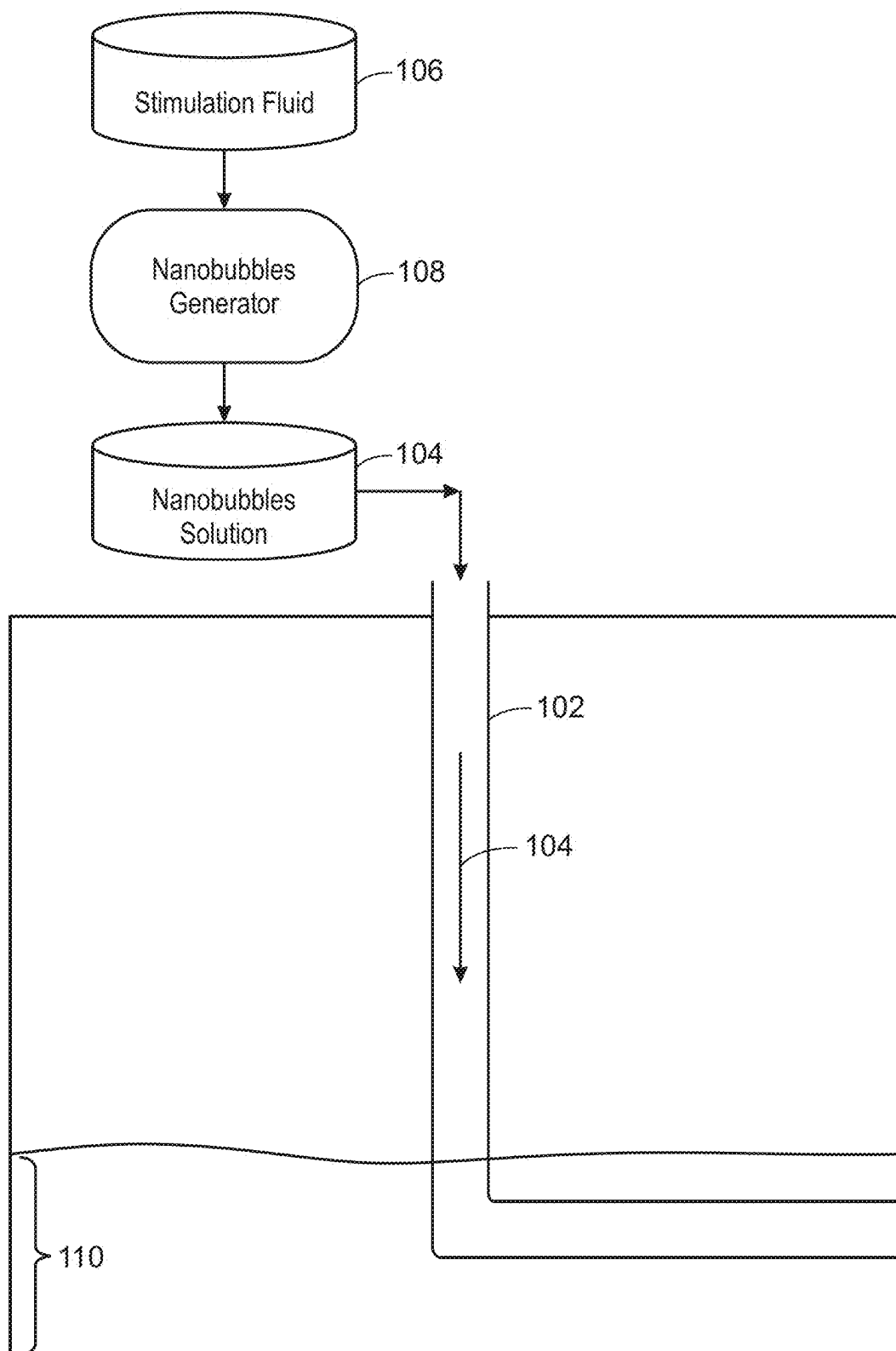


FIG. 1

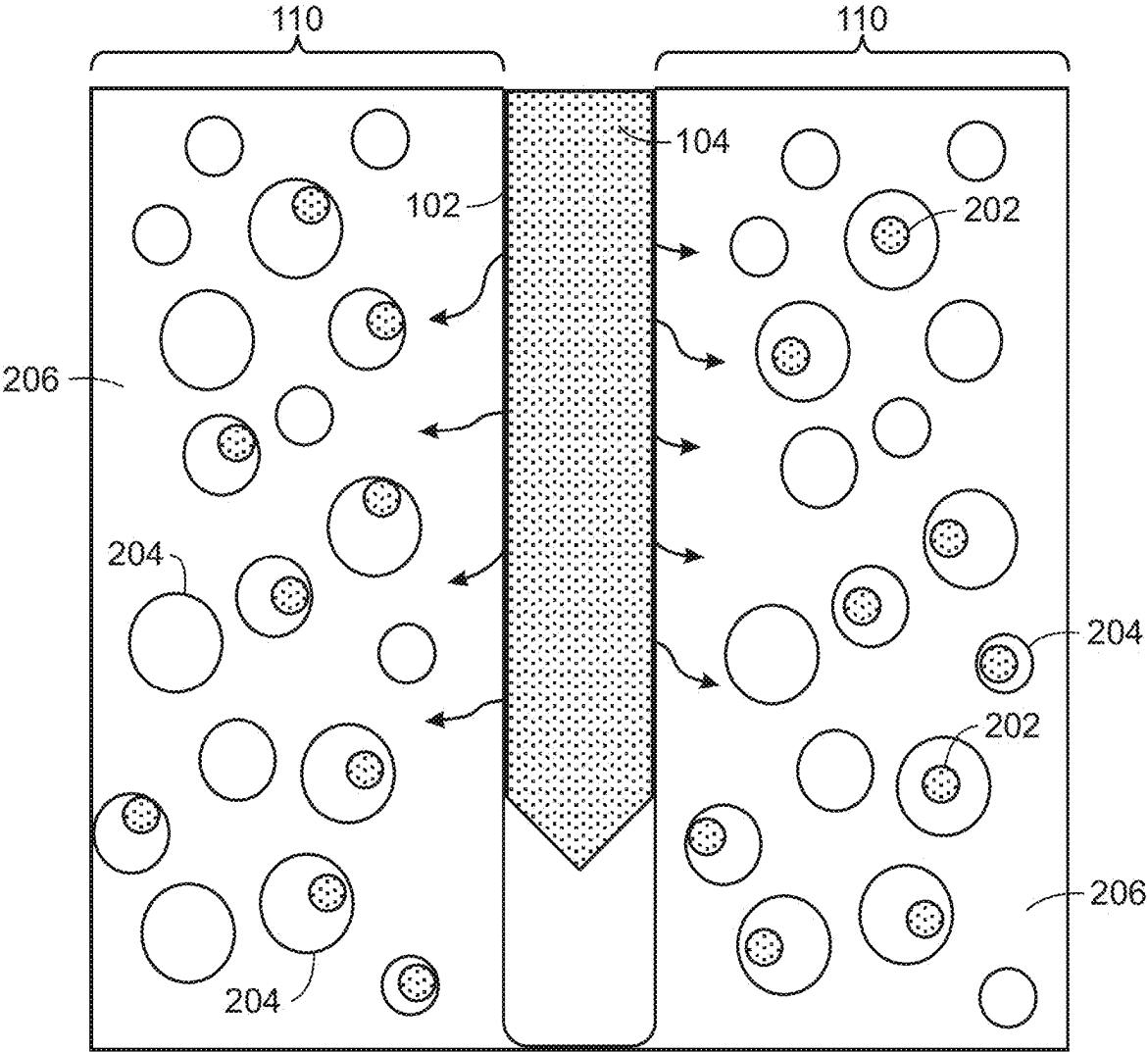


FIG. 2

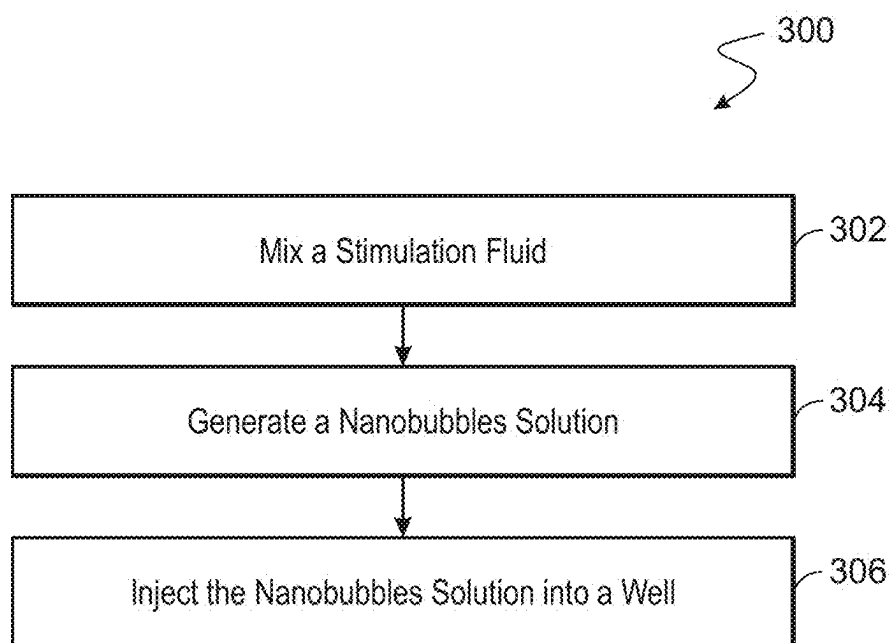


FIG. 3

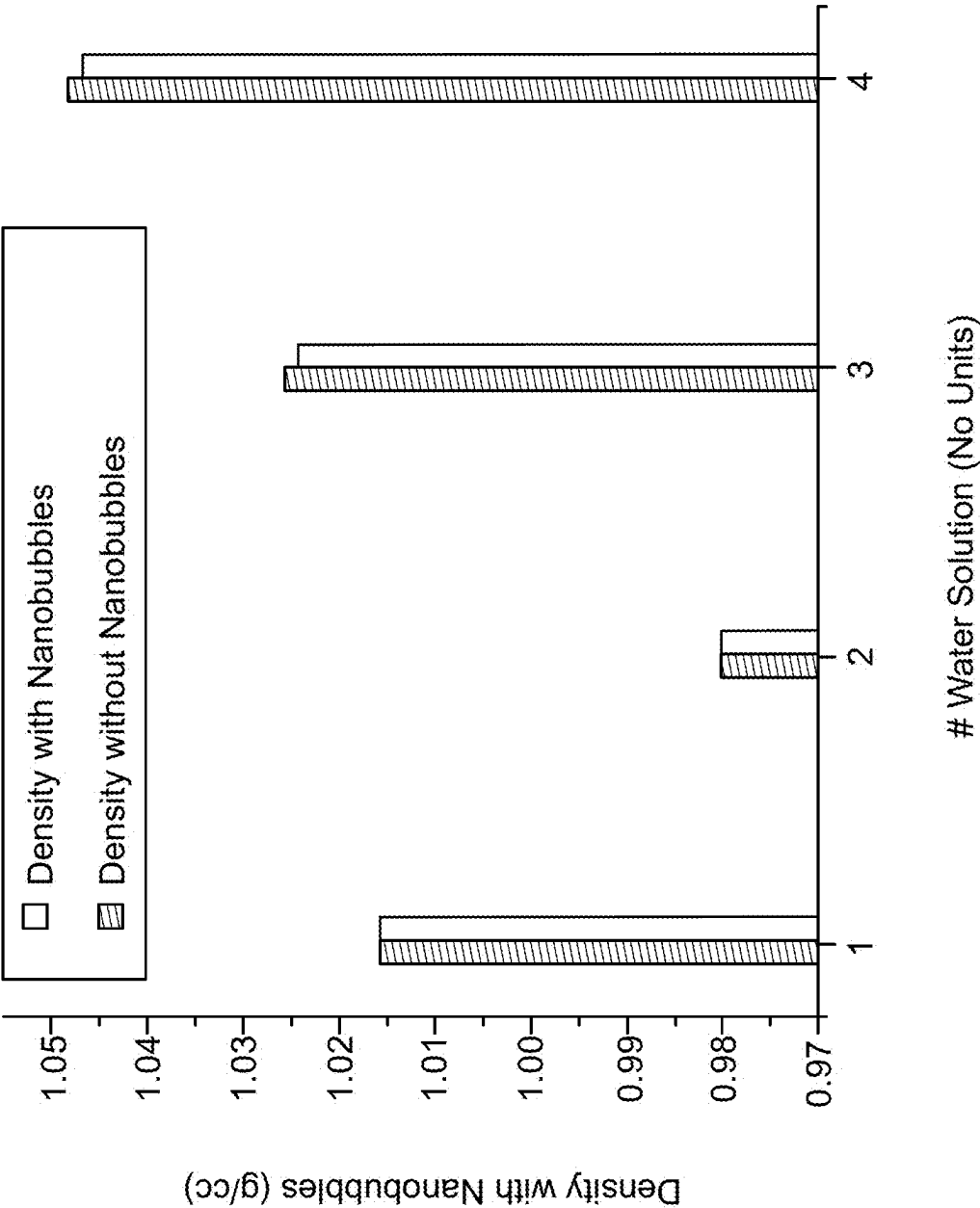


FIG. 4

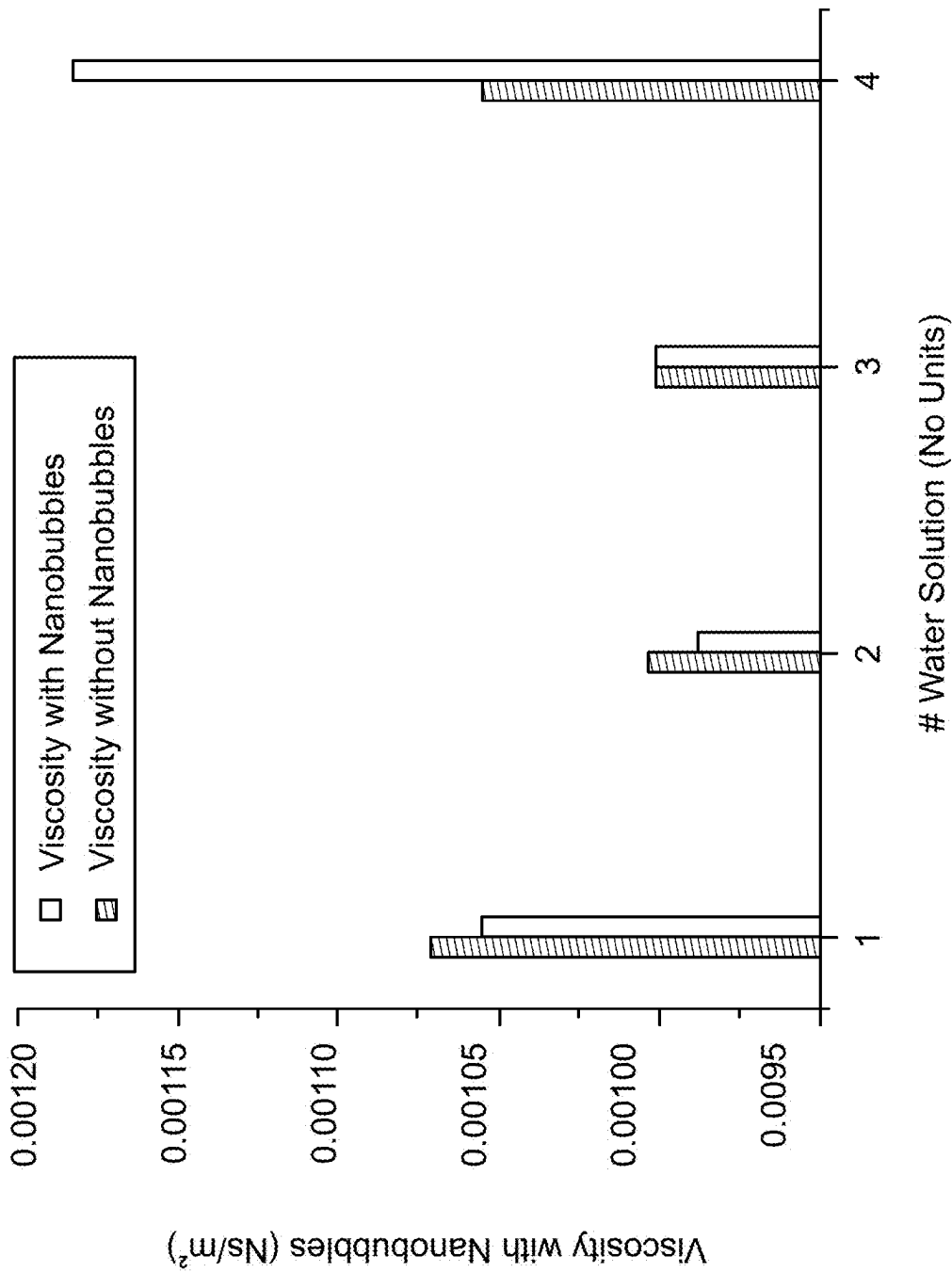


FIG. 5

METHOD FOR WELL STIMULATION USING NANOBUBBLES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a divisional and claims the benefit of priority of U.S. patent application Ser. No. 18/464,455, filed Sep. 11, 2023, the entire contents of which are incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates to methods of enhancing well performance in subsurface formations.

BACKGROUND

[0003] Acid stimulation is an effective method to enhance well performance in subsurface formations. In acid stimulation, an acid-based fluid such as HCl would be typically injected at various concentrations to create conductive channels to enhance the flow paths for hydrocarbons. The acid stimulation fluid that is used typically involves various components depending on the desired treatment. These include a corrosion inhibitor, a surfactant, one or more types of acids, and others.

[0004] The use of CO₂ in acid treatment jobs is limited although CO₂ offers some advantages. For instance, CO₂ is an efficient solvent for removing formation damage and condensate banking around the wellbore. This is attributed to the CO₂ being miscible with the condensate. The use of CO₂ in the acid stimulation fluid will also increase the amount of CO₂ avoided, thereby contributing to reducing the carbon footprint of the operation.

SUMMARY

[0005] An embodiment described herein provides a method for stimulating a well with nanobubbles. The method includes obtaining a stimulation fluid and generating a nanobubbles solution, wherein the nanobubbles solution includes nano-sized bubbles in the stimulation fluid. The nanobubbles solution is injected into the well.

[0006] Another embodiment described herein provides a composition for stimulating an oil well including nano-sized bubbles in a stimulation fluid.

[0007] Another embodiment described herein provides a manufacturing a nanobubbles fluid for stimulating an oil well with nanobubbles. The method includes mixing components to form a stimulation fluid and generating nanobubbles in the stimulation fluid to create a nanobubbles fluid.

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. 1 is a schematic drawing of a process of stimulating a well with a nanobubbles solution.

[0009] FIG. 2 is a schematic drawing of the nanobubbles solution flowing through the well and the CO₂ nanobubbles penetrating the pores of the surrounding rock formation.

[0010] FIG. 3 is a process flow diagram of a method for stimulating a well with a nanobubbles solution.

[0011] FIG. 4 is a plot of the density of water solutions under differing conditions with and without nanobubbles.

[0012] FIG. 5 is a plot of the viscosity of water solutions under differing conditions with and without nanobubbles.

DETAILED DESCRIPTION

[0013] Embodiments described herein use a stimulation fluid as the base fluid for making a nanobubble solution for the stimulation treatment. As used herein, nanobubbles are nano-sized gas bubbles, for example, having a size of less than about 1 micrometer (μm), or between about 50 nm and about 700 nm, or between about 100 nm and about 500 nm. Microbubbles have a size between about 1 μm and about 100 μm. Macro-bubbles have a size of greater than about 100 μm. The nanobubble solution is typically made by directly generating the nanobubbles in the stimulation fluid or by combining a solution containing the nanobubbles with the stimulation fluid. In various embodiments, the nanobubbles are formed from O₂, CO₂, N₂, air, or combinations thereof.

[0014] As opposed to nanobubbles, microbubbles, and macro-bubbles have short lifespans in aqueous solutions. They tend to rise quickly to the surface and/or dissolve rapidly. By comparison, due to their small size, nanobubbles stay suspended in solution for an extended period of time, for example, ranging from several hours to several months. The increased stability of nanobubbles give the nanobubble solution a longer lifespan. Further, the nanobubbles will have a larger surface area for the same volume of gas. Currently, nanobubbles are used in multiple industrial applications such as agriculture, aquaculture, wastewater treatment, food processing, cleaning and sterilization, cooling, or extraction, among others.

[0015] FIG. 1 is a schematic drawing of a process 100 of stimulating a well 102 with a nanobubbles solution 104. As shown in the process 100, a stimulation fluid 106 is passed through a nanobubbles generator 108. In various embodiments, the stimulation fluid 106 includes several components typically used in stimulation fluids, such as an acid, a surfactant, a corrosion inhibitor, and other components. The acid may be an inorganic acid, an organic acid, or an acid generating compound. In various embodiments, the acid is a strong inorganic acid, such as hydrochloric acid, sulfuric acid, nitric acid, or other inorganic acids, or a combination thereof. The organic acid can include acetic acid or formic acid, among others. The acid generating compounds can include esters. The surfactant helps to improve the compatibility of the acid with the formation fluids, break down emulsions, and maintain favorable formation wettability. The surfactant can be a cationic surfactant, and an anionic surfactant, or zwitterionic surfactant, depending on the acid and well conditions. The corrosion inhibitor protects the tubulars and piping from acid corrosion. Other additives may include iron control agents, to help solubilize iron ions lowering the formation of iron scale, and H₂S scavengers to lower precipitation of sulfur compounds.

[0016] In some embodiments, the nanobubbles generator 108 uses an ultrasonic transducer in a sonicator to generate the nanobubbles. For example, the nanobubbles generator 108 can generate the nanobubbles in the stimulation fluid 106 directly, by forcing dissolved gases to come out of solution as the nanobubbles. Further, a gas stream can be added to the stimulation fluid 106 prior to the sonication, which breaks the gas into the nanobubbles. In some embodiments, the nanobubbles are generated in a secondary fluid, which is then added to the stimulation fluid 106. In an embodiment, CO₂ is used as the gas to generate the nanobubble solution. The use of CO₂ may provide additional benefits over other gases due to the formation of carbonic acid. The extra energy that the nanobubbles possess makes

the nanobubbles solution **104** more effective, for example, delivering the energy upon bursting or coalescing with other bubbles.

[0017] The nanobubbles solution **104**, made using the stimulation fluid **106** as the base fluid, will enhance the efficiency of well stimulation jobs. The CO₂ nanobubbles will not increase the viscosity of the nanobubbles solution **104** versus the stimulation fluid **106**, thus it can be injected into a well **102** under the same conditions as the stimulation fluid **106**. In addition to the solvency efficiency of CO₂, the CO₂ nanobubbles can deliver additional thermal and mechanical energies downhole and in the near wellbore region of the formation **110** that enhance the removal of formation damage, for example, due to condensate blockage, as described herein.

[0018] FIG. 2 is a schematic drawing of the nanobubbles solution **104** flowing through the well **102** and the CO₂ nanobubbles **202** penetrating the pores **204** of the surrounding rock formation **206**, for example, in the formation **110**. Like numbered items are as described with respect to FIG. 1. The size of the CO₂ nanobubbles **202** allows them to penetrate porous tool for that are generally not accessible with other fluids. Thus, they can interact with resident fluids in these pores **204**. For example, the CO₂ is miscible with intermediate or low carbon oils.

[0019] In gas reservoir applications, condensate blockage is a frequent problem in which the pressure of the reservoir drops below the dew point of the condensate in the natural gas, allowing the condensate to condense out of the gas phase and form a liquid phase. The condensate typically hinders the production of the well **102**, and may kill the well **102**, making it unable to flow. The miscibility of CO₂ with the condensate, would increase the ability of the condensate to flow and free the well from the condensate blockage.

[0020] Further, the CO₂ is a good solvent at increased pressure and temperature conditions. Thus, the CO₂ nanobubbles **202** added to the stimulation fluid will enhance the effectiveness of the stimulation fluid, enabling it to clear any obstructions in the formation **110** near the well **102**.

[0021] FIG. 3 is a process flow diagram of a method **300** for stimulating a well with a nanobubbles solution. The method begins at block **302**, with the mixing of a stimulation fluid. The stimulation fluid may be aqueous based, for example, mixed in a production brine or a production brine with adjustment of ionic content. In other examples, the stimulation fluid may use a simulated brine as the base fluid. The stimulation fluid may be based on oil-in-water emulsions or water-in-oil emulsions.

[0022] The stimulation fluid is prepared by adding an acid, as described herein, to the base fluid. A surfactant is added to the stimulation fluid, before or after the acid. Then, a corrosion inhibitor is added to form the final stimulation fluid.

[0023] At block **304**, a nanobubbles solution is generated. As described herein, the nanobubbles solution may be generated by direct sonication of the stimulation fluid, for example, by passing it over an ultrasonic transducer to force dissolved gases, such as CO₂, to be released and formed the nanobubbles. A gas stream may be added to the stimulation fluid just before the stimulation fluid is passed over the ultrasonic transducer, generating the nanobubbles solution. Further, a nanobubble fluid can be separately generated, for

example, using the same base fluid as the stimulation fluid, then added to the stimulation fluid to generate the nanobubbles solution.

[0024] At block **306**, the nanobubbles solution is injected into a well. The viscosity and density are substantially the same as the stimulation fluid, thus, the injection conditions are similar. The viscosity and density of the nanobubbles solution is discussed further with respect to FIGS. 4 and 5.

[0025] FIG. 4 is a plot of the density of different types of water solutions with and without nanobubbles. In FIG. 4, water solution **1** is tap water, **2** is distilled water, **3** is a slurry water, and **4** is seawater. As this plot shows, the nanobubble solutions are generally similar in density to the base fluid. These results indicate that the nanobubbles solution will not exert extra hydraulic pressure on the formation.

[0026] FIG. 5 is a plot of the viscosity of water solutions under differing conditions with and without nanobubbles. As for FIG. 4, in FIG. 5, water solution **1** is tap water, **2** is distilled water, **3** is a slurry water, and **4** is seawater. As this plot shows, the viscosity of nanobubble solutions are typically similar to that of the base fluid. In contrast, other additives, such as polymers, may substantially increase the viscosity of the base fluid. Further, the comparable viscosity to the stimulation fluid indicates that the nanobubbles solution will be easy to flow back once the well is put on production.

EMBODIMENTS

[0027] An embodiment described herein provides a method for stimulating a well with nanobubbles. The method includes obtaining a stimulation fluid and generating a nanobubbles solution, wherein the nanobubbles solution includes nano-sized bubbles in the stimulation fluid. The nanobubbles solution is injected into the well.

[0028] In an aspect, combinable with any other aspect, the method includes mixing the stimulation fluid. Mixing the stimulation is performed by mixing an acid into a base fluid to form an acid solution, adding a surfactant to the acid solution to form an acid/surfactant solution, and adding a corrosion inhibitor to the acid/surfactant solution to form the stimulation fluid. In an aspect, the acid includes an inorganic acid, an organic acid, or an acid generating compound. In an aspect, the acid includes hydrochloric acid, sulfuric acid, or nitric acid, or any combination thereof. In an aspect, the acid includes acetic acid or formic acid.

[0029] In an aspect, combinable with any other aspect, the method includes generating the nano-sized bubbles. In an aspect, generating the nano-sized bubbles includes passing the stimulation fluid including a dissolved gas through a sonicator to generate the nanobubbles solution. In an aspect, generating the nano-sized bubbles includes injecting a gas into the stimulation fluid to form gas bubbles in the stimulation fluid and passing the stimulation fluid with the gas bubbles through a sonicator to generate the nanobubbles solution. In an aspect, generating the nano-sized bubbles includes dissolving a gas in a fluid, passing the fluid including the dissolved gas creating a nanobubbles fluid, and mixing the nanobubbles fluid with the stimulation fluid to create the nanobubbles solution.

[0030] In an aspect, combinable with any other aspect, the nano-sized bubbles remain suspended in the nanobubbles solution for more than about 60 minutes.

[0031] Another embodiment described herein provides a composition for stimulating an oil well including nano-sized bubbles in a stimulation fluid.

[0032] In an aspect, combinable with any other aspect, the stimulation fluid includes an acid, a surfactant, and a corrosion inhibitor.

[0033] In an aspect, combinable with any other aspect, the acid includes an inorganic acid, an organic acid, or an acid generating compound.

[0034] In an aspect, combinable with any other aspect, the acid includes hydrochloric acid, sulfuric acid, or nitric acid, or any combination thereof.

[0035] In an aspect, combinable with any other aspect, the acid includes acetic acid or formic acid.

[0036] In an aspect, combinable with any other aspect, the nano-sized bubbles include CO₂.

[0037] In an aspect, combinable with any other aspect, a density of the stimulation fluid with the nano-sized bubbles is within 0.01 g/cc of the stimulation fluid without the nano-sized bubbles.

[0038] In an aspect, combinable with any other aspect, the nano-sized bubbles have a higher miscibility with intermediate and low carbon oils than the stimulation fluid.

[0039] Another embodiment described herein provides a manufacturing a nanobubbles fluid for stimulating an oil well with nanobubbles. The method includes mixing components to form a stimulation fluid and generating nanobubbles in the stimulation fluid to create a nanobubbles fluid.

[0040] In an aspect, the method includes generating nano-sized bubbles in an aqueous liquid; and combining the aqueous liquid with the stimulation fluid.

[0041] In an aspect, the method includes generating nano-sized bubbles in the stimulation fluid by sonication.

[0042] In an aspect, combinable with any other aspect, the method includes mixing the aqueous liquid with stimulation fluid at the well site to create the nanobubbles fluid. Other implementations are also within the scope of the following claims.

What is claimed is:

1. A nanobubbles solution for stimulating an oil well, wherein the nanobubbles solution is prepared by:

mixing an acid into a base fluid to form an acid solution; adding a surfactant to the acid solution to form an acid/surfactant solution;

adding a corrosion inhibitor to the acid/surfactant solution to form a stimulation fluid;

generating nano-sized bubbles in the stimulation fluid by a nanobubble generator to form a nanobubbles solution, wherein the nanobubbles solution comprises nano-sized bubbles and has a viscosity as that of the base fluid.

2. The nanobubbles solution of claim 1, wherein the acid comprises an inorganic acid, an organic acid, or an acid generating compound.

3. The nanobubbles solution of claim 2, wherein the acid comprises hydrochloric acid, sulfuric acid, or nitric acid, or any combination thereof.

4. The nanobubbles solution of claim 2, wherein the acid comprises acetic acid or formic acid.

5. The nanobubbles solution of claim 1, wherein the nano-sized bubbles are generated by sonication.

6. The nanobubbles solution of claim 1, wherein the nano-sized bubbles comprise CO₂.

7. The nanobubbles solution of claim 1, wherein a density of the nanobubbles solution comprising the nano-sized bubbles is within 0.01 g/cc of the nanobubbles solution without the nano-sized bubbles.

8. The nanobubbles solution of claim 1, wherein the nano-sized bubbles in the nanobubbles solution have a higher miscibility with intermediate and low carbon oils when compared to the nanobubbles solution without the nano-sized bubbles.

9. A composition for stimulating an oil well, the composition comprising:

an acid selected from an inorganic acid, an organic acid, or an acid generating compound;

a surfactant selected from a cationic surfactant, an anionic surfactant, or a zwitterionic surfactant;

a corrosion inhibitor; and

nano-sized bubbles.

10. The composition of claim 9, wherein the acid comprises hydrochloric acid, sulfuric acid, nitric acid, or any combination thereof.

11. The composition of claim 9, wherein the acid comprises acetic acid or formic acid.

12. The composition of claim 9, wherein the nano-sized bubbles comprise CO₂.

13. The composition of claim 9, wherein a density of the composition comprising the nano-sized bubbles is within 0.01 g/cc of the composition without the nano-sized bubbles.

14. The composition of claim 9, wherein the nano-sized bubbles in the composition have a higher miscibility with intermediate and low carbon oils when compared to the composition without the nano-sized bubbles.

15. A method of manufacturing a nanobubbles fluid for stimulating an oil well with nanobubbles, the method comprising:

mixing an acid with a base fluid to form an acid mixture; adding a surfactant and a corrosion inhibitor to the acid mixture to form a stimulation fluid; and

generating nano-sized bubbles in the stimulation fluid by a nanobubble generator to form a nanobubbles fluid, wherein the nanobubbles fluid has a viscosity as that of the base fluid.

16. The method of claim 15, comprising:

generating nano-sized bubbles in an aqueous liquid; and combining the aqueous liquid with the stimulation fluid.

17. The method of claim 15, comprising generating nano-sized bubbles in the stimulation fluid by sonication.

18. The method of claim 17, wherein the nano-sized bubbles comprise CO₂.

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